Inspecting failures and anomalies

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Teach the evidence-to-conclusion chain for Inspecting failures and anomalies

Separate valid slow behavior, demonstrated measurement errors and untested explanations.

Keep one habit: Model the main case → reveal the picture gradually → probe limits → assess independent reasoning → plan a changed-case return.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of inspecting failures and anomalies

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of running the primary experiment

Reference example, not the only acceptable wording: Teacher explanation: The primary experiment follows the frozen plan and captures every eligible outcome. Unexpected failures are recorded rather than silently retried away.

Run declared cases, store raw paired results and deviations, and keep primary analysis separate from exploratory repairs. Stop or escalate only according to the stated rule.

Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of power and resource tradeoffs

Reference example, not the only acceptable wording: Teacher explanation: Power concerns detecting a declared effect under a specified model and test. Planning depends on effect size, variability, unit independence and budget.

Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of privacy and data governance

Reference example, not the only acceptable wording: Teacher explanation: Data governance specifies who may use which records, for what purpose and for how long. Coded data can still be reidentifiable.

For fictional records separate participant-code table from analysis values; limit access and retention. Remove unnecessary fields and small-group details before a permitted aggregate export.

Pictured end goal

Explain inspecting failures and anomalies using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Running the primary experiment

Power and resource tradeoffs

Privacy and data governance

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. Failure inspection generates explanations to test. An unusual result is not automatically a bad data point. Use the supplied constructed scenarios to make the **mechanism** inspectable.

Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of inspecting failures and anomalies

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model the whole case and check each inference

> Keep the independent answer private until after the first attempt.

Definitions and complete contract

Failure inspection generates explanations to test. An unusual result is not automatically a bad data point.

Compare raw **output**s, **input** validity and procedure traces. Correct demonstrable measurement errors with provenance; retain valid unusual **observation**s. Label post-result investigations exploratory.

Anomaly item	Complete supplied contract
Main valid durations	1, 2, 2, 15 seconds; all stipulated valid traces
Proposed main deletion	Remove fifteen seconds without a rule
Main analysis	Retain all four in the declared primary summary
Separate picture durations	2, 2, 2, 10 seconds
Picture log	Fourth run records a source timeout; plan includes timeouts
Correction requirement	Evidence of a measurement error and preserved provenance, not dislike of an outcome
Scope	Supplied timing traces; no actual timing, timeout generation or root-cause experiment

Main evidence ledger and checked key

Main calculation	Complete arithmetic	Interpretation
Retained durations	1, 2, 2, 15 seconds	All valid under supplied contract
Sum / count	20 seconds / 4	No undeclared exclusion

Main calculation	Complete arithmetic	Interpretation
Mean	20/4 = 5 seconds	Sensitive to the long valid duration
Median	Middle values 2 and 2; average 2 seconds	Different descriptive aspect
Wrong-deletion illustration	(1+2+2)/3 = 5/3 seconds	Shows distortion; not a replacement primary result
Cause of fifteen-second duration	Not supplied	Propose an investigation without inventing its outcome

Main 1, 2, 2, 15 gives sum twenty, mean five and median two seconds. All traces are valid; undeclared deletion of fifteen is not justified. Separate picture 2, 2, 2, 10 gives mean four and median two, with timeout included by the plan. Deleting ten just for appearance would conceal relevant behavior. The timeout event does not prove every root cause.

Independent 2, 3, 4, 20 gives mean 29/4 = 7.25 and median (3+4)/2 = 3.5 seconds. Retain twenty and propose an honestly labelled exploratory trace investigation. Genuine measurement corrections need evidence/provenance; no actual timings or causal test occur in this calculation.

Inspecting failures and anomalies — evidence trace

INPUT AND SCOPE

Durations 1, 2, 2, 15 seconds are all valid traces. Team deletes 15 as an outlier without a rule.

CHECK OR CALCULATION

Compare raw outputs, input validity and procedure traces. Correct demonstrable measurement errors with provenance; retain valid unusual observations.

CONCLUSION AND LIMIT

The slow valid case must remain in the declared primary analysis. Investigate mechanism separately; mean 5 and median 2 reveal different aspects.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: The slow valid case must remain in the declared primary analysis. Investigate **mechanism** separately; mean 5 and median 2 reveal different aspects. Qualification: Post-hoc exclusion based on an unwanted outcome can distort the conclusion. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Durations 1, 2, 2, 15 seconds are all valid traces. Team deletes 15 as an outlier without a rule.

The slow valid case must remain in the declared primary analysis. Investigate **mechanism** separately; mean 5 and median 2 reveal different aspects.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Boundary and misconception probes

Anomaly audit	Checked interpretation
Value unusually large	Not automatically invalid data
All traces stipulated valid	Keep them under the declared inclusion rule
Mean differs from median	Different summaries; neither licenses deleting the tail
Log shows timeout	Event evidence, not a complete causal explanation
Known unit conversion error demonstrated	Correct transparently with original record and evidence
No evidence of capture error	Do not invent one to improve results
Follow-up inspired by outcomes	Label exploratory and preserve original analysis
Wrong-deletion calculation shown	Illustration of distortion, not a new approved result

Separate pictured facts and full reasoning

Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration.

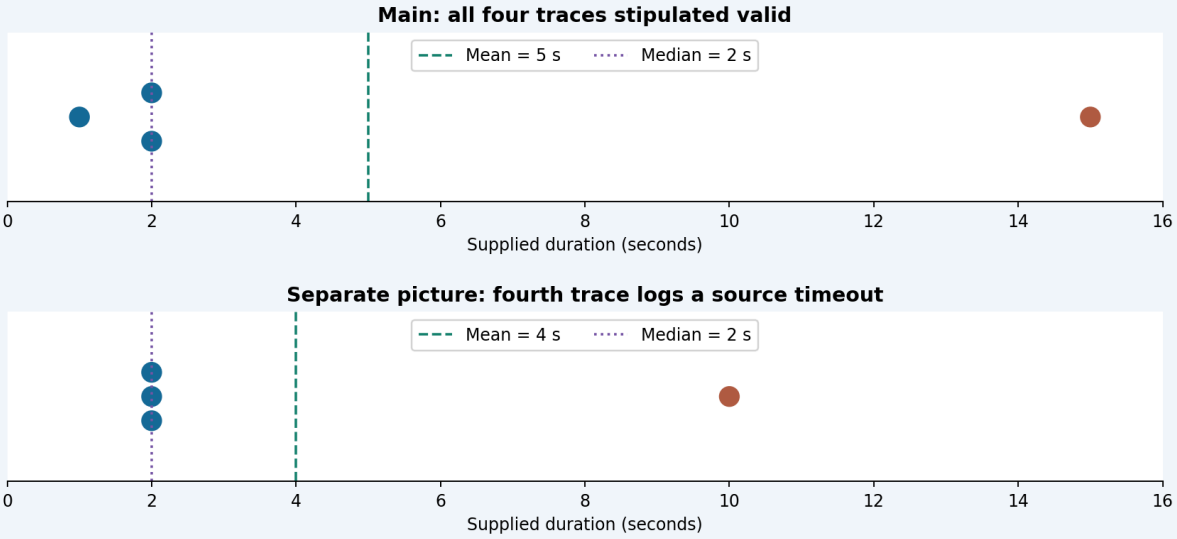
Durations are 2,2,2,10 seconds. Logs show a source timeout on the fourth run. The plan includes timeouts in reporting.

The separate picture records 2, 2, 2 and 10 seconds. The sum is sixteen, mean four and median two seconds. The fourth log records a source timeout, and the predefined reporting plan includes timeouts. That record remains in the primary summary. Deleting it merely because it looks bad would produce a two-second mean and conceal the stipulated slow case. A timeout event does not identify every underlying cause.

Inspect trace boundaries, source behavior and known conditions, then design a controlled follow-up; label proposed explanations and follow-ups as untested until evidence exists.

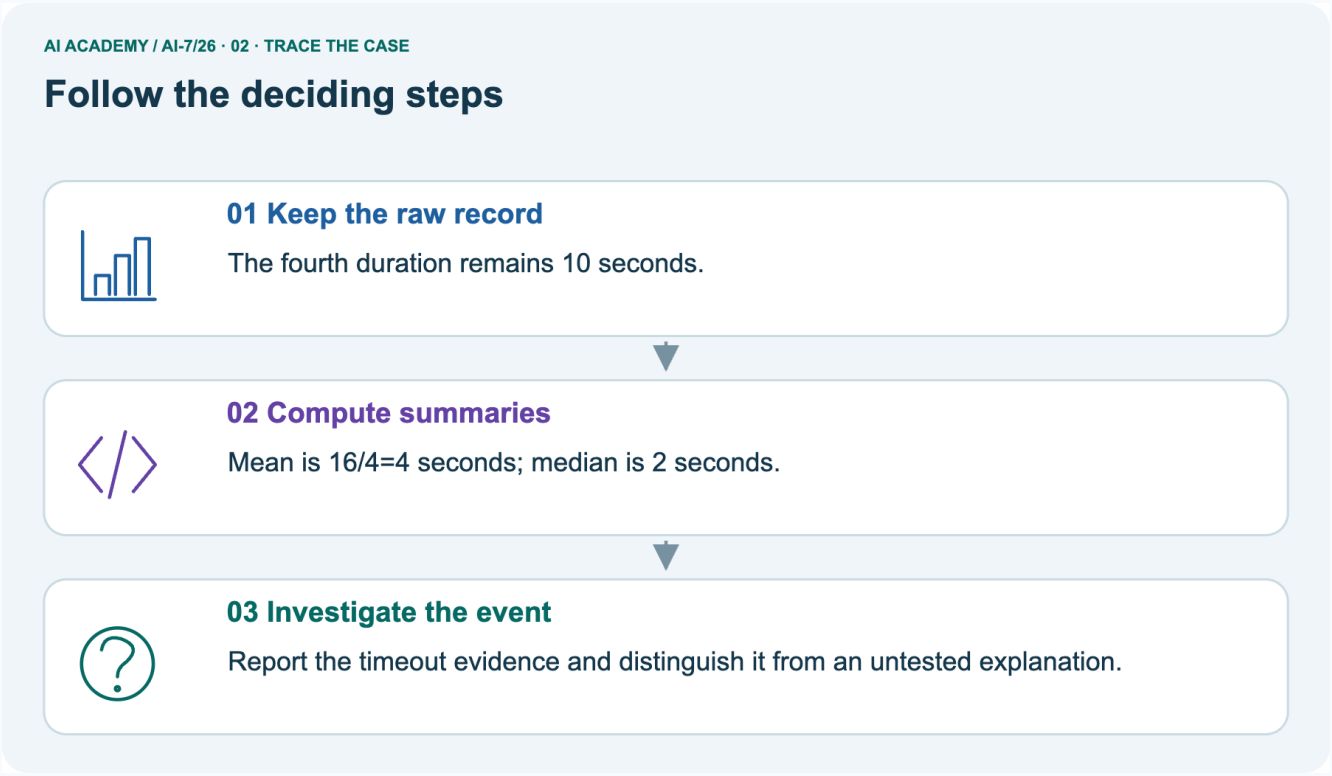
Separate picture quantity	Supported record
Durations	2, 2, 2, 10 seconds
Mean	16/4 = 4 seconds
Median	(2+2)/2 = 2 seconds
Fourth log	Source timeout recorded
Plan	Includes timeouts in reporting
Delete ten because it looks bad	Violates plan; hypothetical mean falls to two
Root cause	Not fully identified by timeout event
Follow-up status	Proposed exploratory investigation, not a completed test

Investigate a slow valid case; keep it in the planned analysis



Vertical offsets separate overlapping dots; they carry no measurement. No actual timings were run.
The timeout event does not establish every underlying cause; a proposed explanation still needs a test.

Picture trace and changed-condition key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep the raw record	The fourth duration remains 10 seconds.
2. Compute summaries	Mean is 16/4=4 seconds; median is 2 seconds.
3. Investigate the event	Report the timeout evidence and distinguish it from an untested explanation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An **anomaly** invites investigation, not automatic deletion. A slow failure may be part of real service behaviour; the predefined rule determines inclusion.

A logged timeout does not identify every underlying cause; report conditions and test explanations.

Change: Delete the 10-second record only because it looks bad.

Prediction: The mean becomes 2 seconds but the selective report violates this plan.

Reason: Observed inconvenient outcomes are not an approved exclusion reason.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Anomaly	An unusual observation requiring investigation
Provenance-backed correction	A repair justified by source evidence
Exploratory analysis	A follow-up chosen after results

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

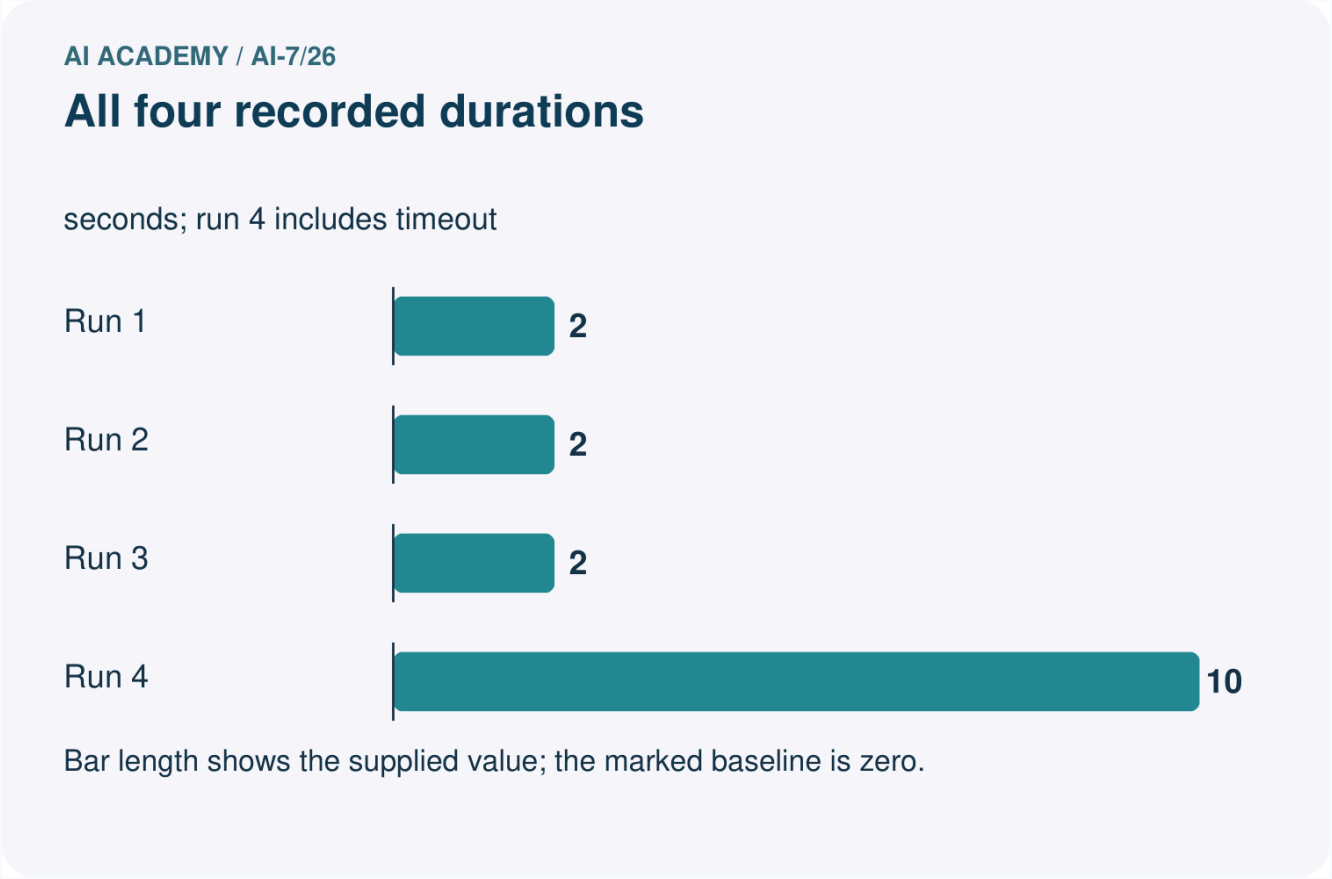
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Inspecting failures and anomalies

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/26

Inspecting failures and anomalies

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep the raw record

The fourth duration remains 10 seconds.



2. Compute summaries

Mean is 16/4=4 seconds; median is 2 seconds.



3. Investigate the event

Report the timeout evidence and distinguish it from an untested explanation.

CHANGE ONE THING

Delete the 10-second record only because it looks bad.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Durations 1, 2, 2, 15 seconds are all valid traces. Team deletes 15 as an outlier without a rule. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: The slow valid case must remain in the declared primary analysis. Investigate **mechanism** separately; mean 5 and median 2 reveal different aspects.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Inspecting failures and anomalies. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Post-hoc exclusion based on an unwanted outcome can distort the conclusion.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?

The fourth duration remains 10 seconds. Mean is 16/4=4 seconds; median is 2 seconds. Report the timeout **evidence** and distinguish it from an untested explanation.

Changed case: The mean becomes 2 seconds but the selective report violates this plan. Observed inconvenient outcomes are not an approved exclusion reason.

Changed-case reasoning: The mean becomes 2 seconds but the selective report violates this plan. Observed inconvenient outcomes are not an approved exclusion reason.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main 1, 2, 2, 15 gives sum twenty, mean five and median two seconds. All traces are valid; undeclared deletion of fifteen is not justified. Separate picture 2, 2, 2, 10 gives mean four and median two, with timeout included by the plan. Deleting ten just for appearance would conceal relevant behavior. The timeout event does not prove every root cause.

Independent 2, 3, 4, 20 gives mean 29/4 = 7.25 and median (3+4)/2 = 3.5 seconds. Retain twenty and propose an honestly labelled exploratory trace investigation. Genuine measurement corrections need evidence/provenance; no actual timings or causal test occur in this calculation.

Diagnose the current misconceptions

The worked result establishes every future case: Post-hoc exclusion based on an unwanted outcome can distort the conclusion.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Values 2, 3, 4, 20 are valid. Compute mean and median and propose an honest investigation.

Teacher explanation: Mean7.25, median3.5. Keep20; inspect its case/trace for a **mechanism** rather than silently deleting it.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Failure inspection generates explanations to test. An unusual result is not automatically a bad data point. Compare raw **output**s, **input** validity and procedure traces.

Correct demonstrable measurement errors with provenance; retain valid unusual **observation**s. Label post-result investigations exploratory.

W2 Trace the worked case

Durations 1, 2, 2, 15 seconds are all valid traces. Team deletes 15 as an outlier without a rule. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: The slow valid case must remain in the declared primary analysis. Investigate **mechanism** separately; mean 5 and median 2 reveal different aspects.

W3 Audit the limit

A report omits this qualification: Post-hoc exclusion based on an unwanted outcome can distort the conclusion. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Post-hoc exclusion based on an unwanted outcome can distort the conclusion.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Post-hoc exclusion based on an unwanted outcome can distort the conclusion.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: The slow valid case must remain in the declared primary analysis. Investigate **mechanism** separately; mean 5 and median 2 reveal different aspects. Qualification: Post-hoc exclusion based on an unwanted outcome can distort the conclusion. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Values 2, 3, 4, 20 are valid. Compute mean and median and propose an honest investigation.

Teacher explanation: Mean7.25, median3.5. Keep20; inspect its case/trace for a **mechanism** rather than silently deleting it.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Mean7.25, median3.5. Keep20; inspect its case/trace for a **mechanism** rather than silently deleting it. Require intermediate reasoning and the stated scope. Fresh retry: One duration recorded2000ms among values in seconds.

Expected reasoning: Convert documented2000ms to2 seconds with provenance; this is a unit correction, not rejection because it looks large.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled inspecting failures and anomalies **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of inspecting failures and anomalies.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Robustness across conditions. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/27, ai-7/29, ai-7/32. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of inspecting failures and anomalies

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Durations 1, 2, 2, 15 seconds are all valid traces. Team deletes 15 as an outlier without a rule. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: The slow valid case must remain in the declared primary analysis. Investigate **mechanism** separately; mean 5 and median 2 reveal different aspects.

Check every learner before discussion: Use the worked case for Inspecting failures and anomalies. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Post-hoc exclusion based on an unwanted outcome can distort the conclusion.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Delete the 10-second record only because it looks bad. Require a **prediction** and a reason before comparing. Expected result: The mean becomes 2 seconds but the selective report violates this plan.

Reason: Observed inconvenient outcomes are not an approved exclusion reason.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

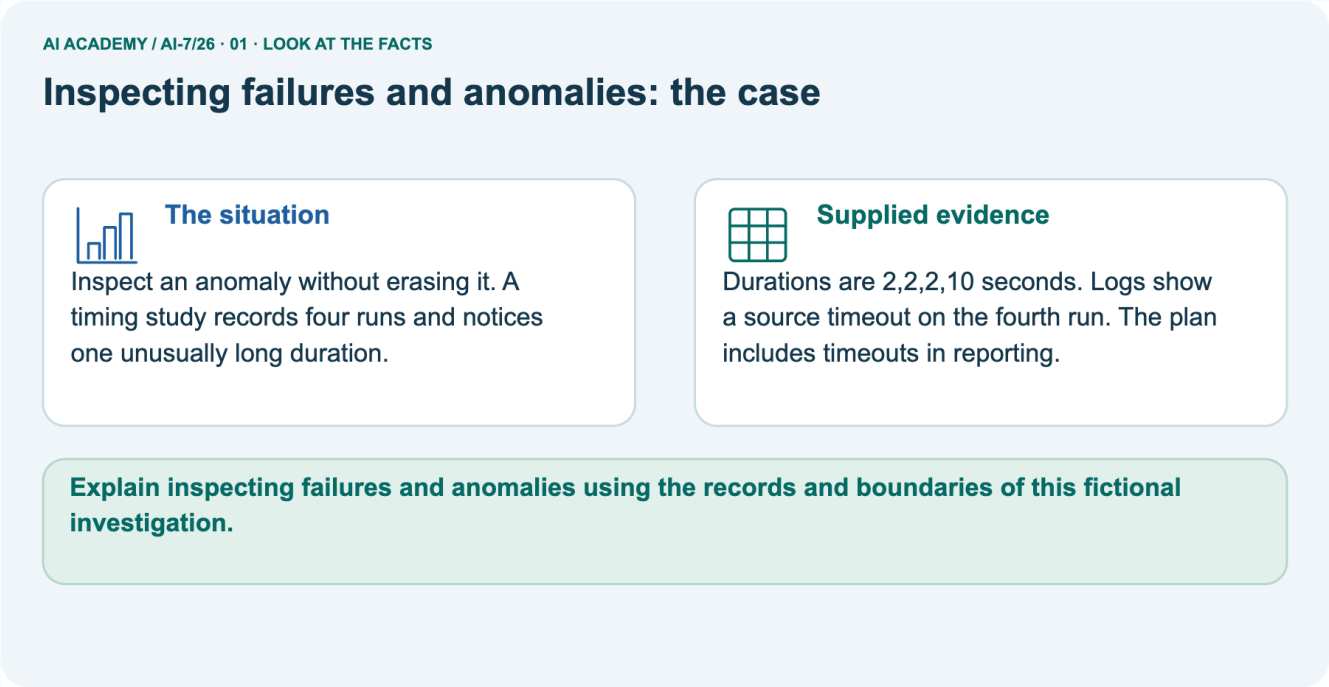
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Inspecting failures and anomalies: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

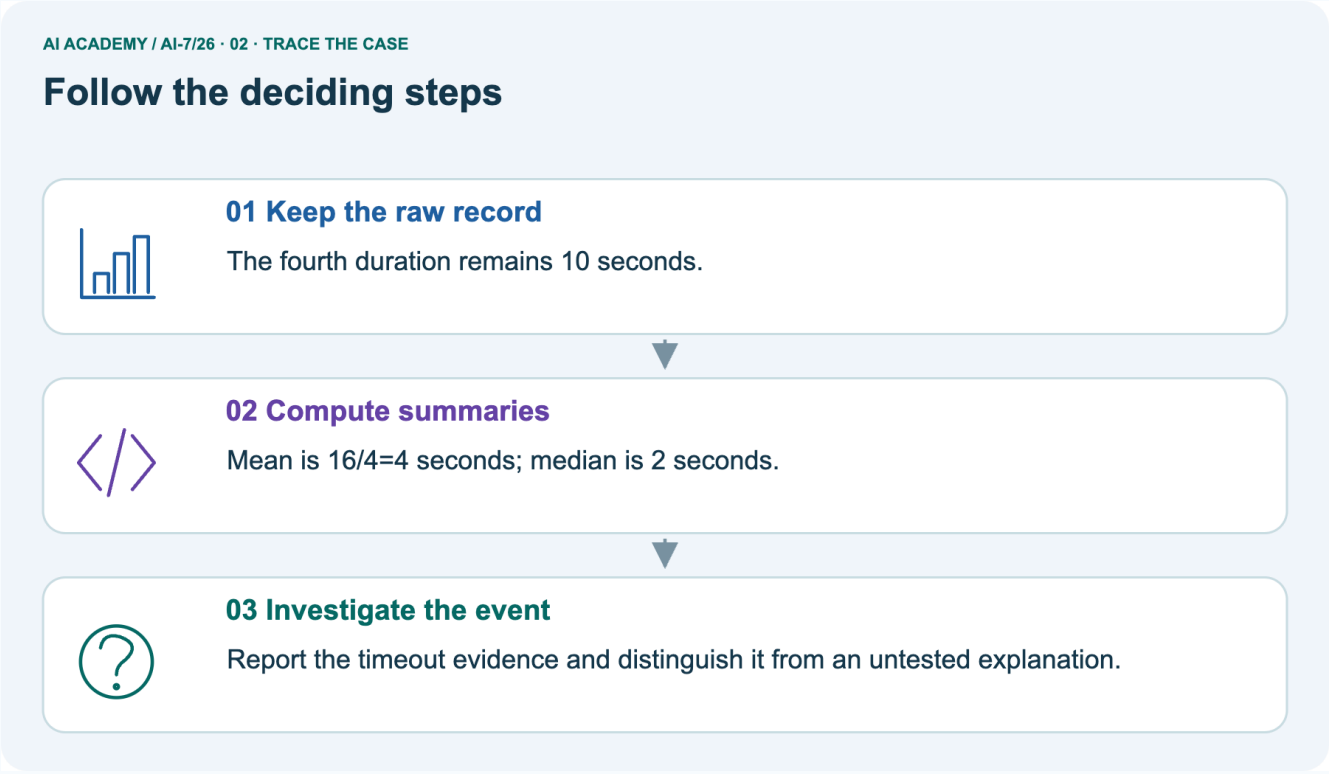
Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration. Supplied facts: Durations are 2,2,2,10 seconds. Logs show a source timeout on the fourth run.

The plan includes timeouts in reporting.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



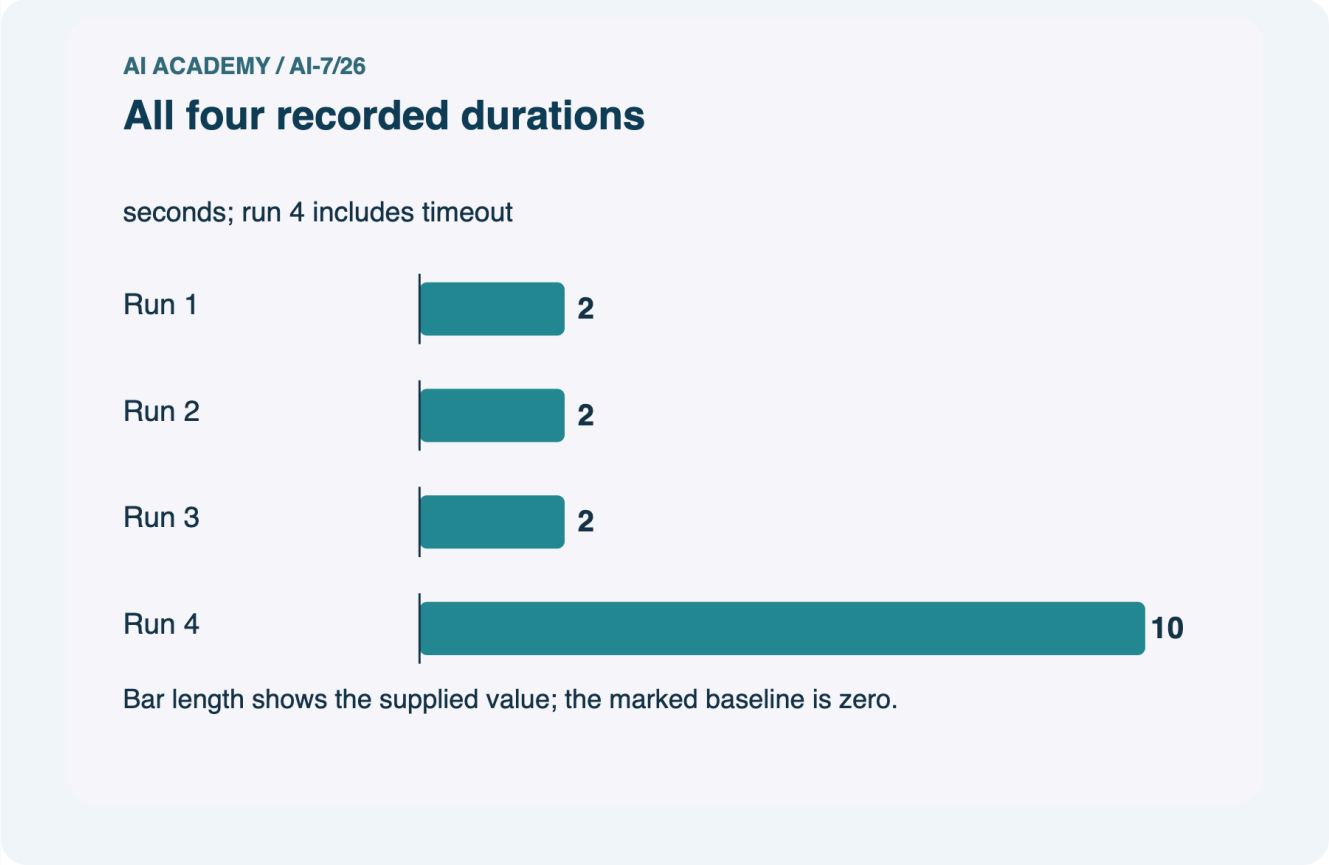
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the raw record: The fourth duration remains 10 seconds. → Compute summaries: Mean is 16/4=4 seconds; median is 2 seconds. → Investigate the event: Report the timeout **evidence** and distinguish it from an untested explanation.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. All four recorded durations



Follow the labelled picture with the matching explanation. Open the vector diagram

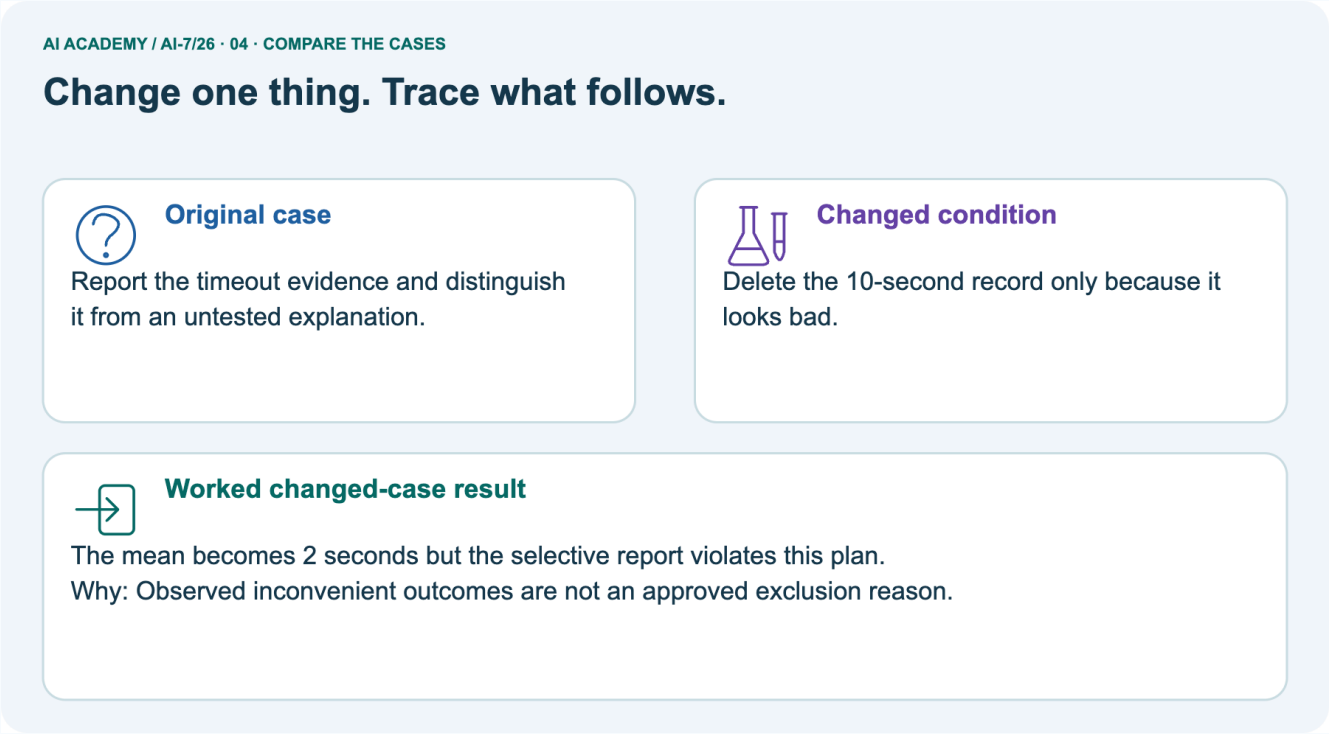
All four recorded durations. Units: seconds; run 4 includes timeout. Run 1: 2. Run 2: 2. Run 3: 2. Run 4: 10.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report the timeout **evidence** and distinguish it from an untested explanation.
Changed condition: Delete the 10-second record only because it looks bad.

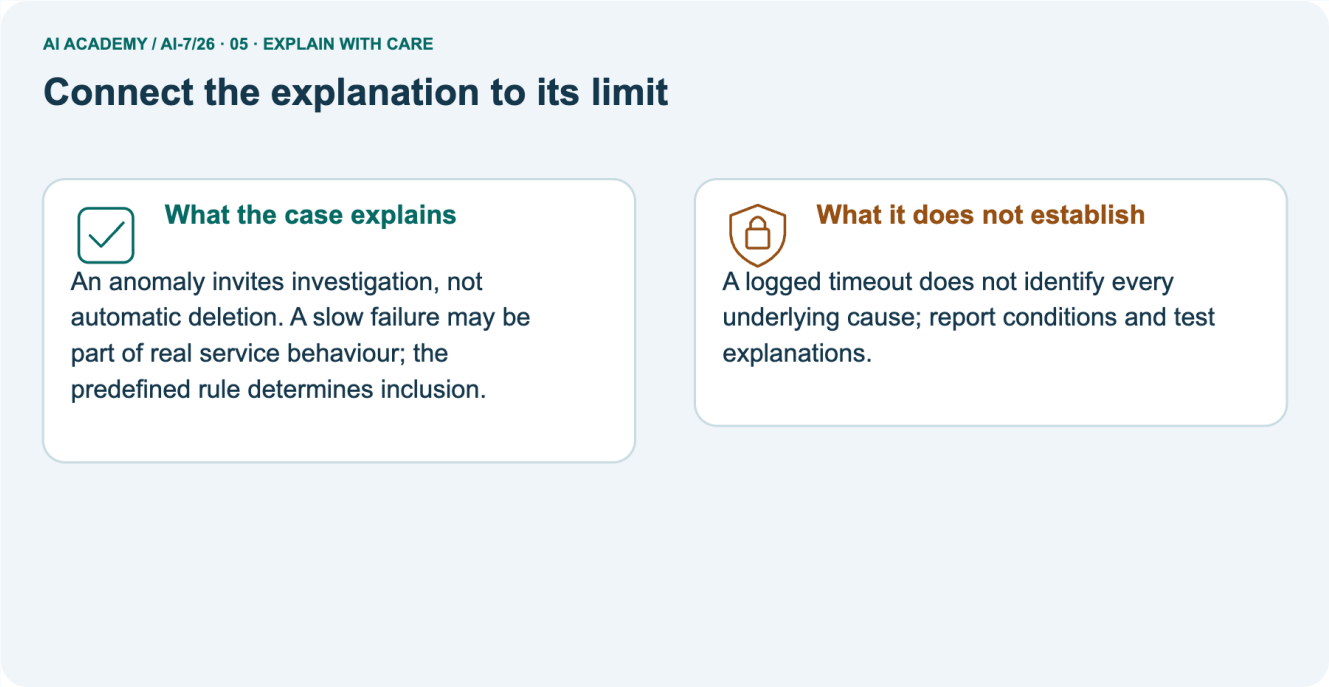
Expected result: The mean becomes 2 seconds but the selective report violates this plan.
Reason: Observed inconvenient outcomes are not an approved exclusion reason.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Observed inconvenient outcomes are not an approved exclusion reason.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An **anomaly** invites investigation, not automatic deletion. A slow failure may be part of real service behaviour; the predefined rule determines inclusion.

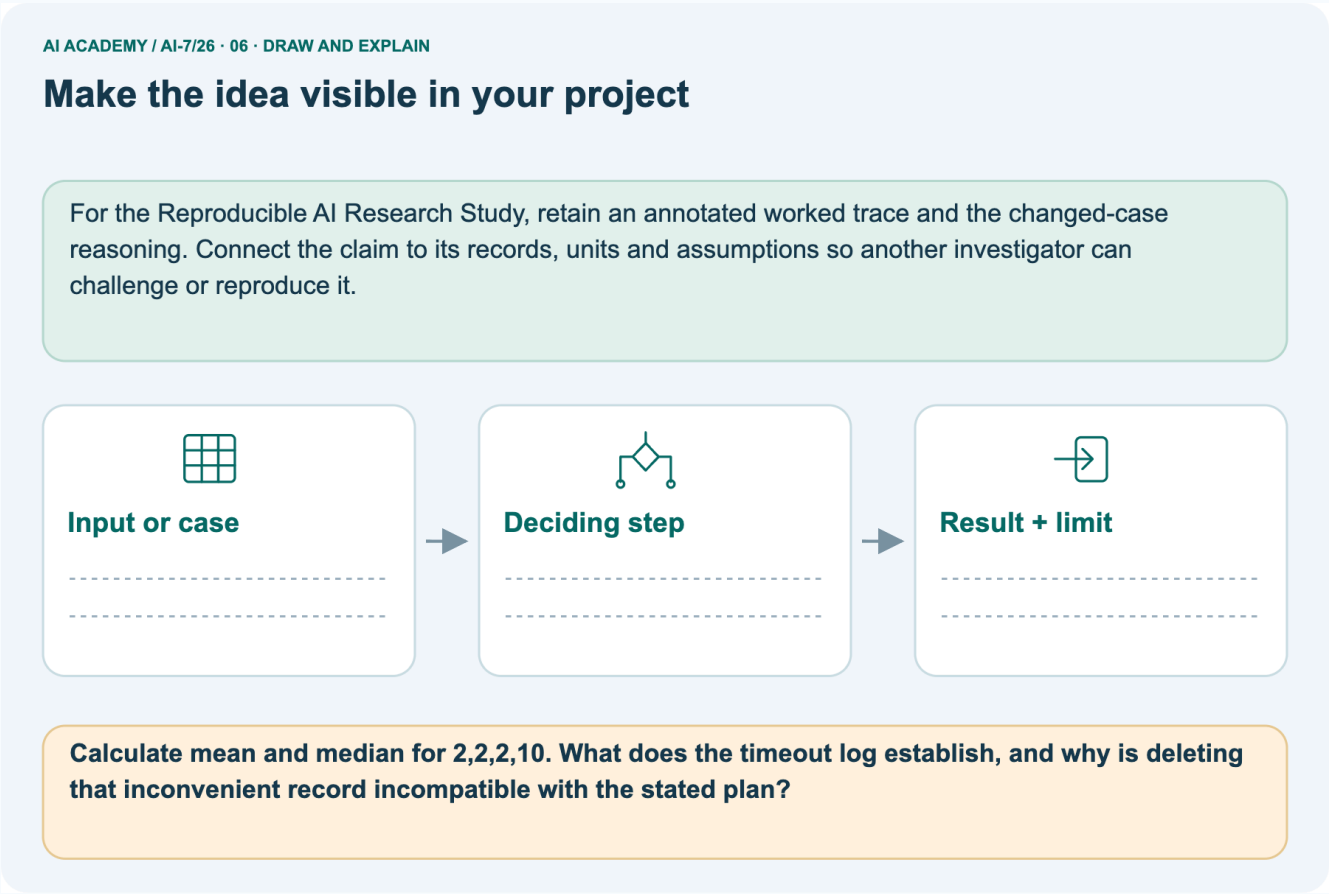
Limit: A logged timeout does not identify every underlying cause; report conditions and test explanations.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A logged timeout does not identify every underlying cause; report conditions and test explanations.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The fourth duration remains 10 seconds. Mean is 16/4=4 seconds; median is 2 seconds. Report the timeout **evidence** and distinguish it from an untested explanation.

Changed case: The mean becomes 2 seconds but the selective report violates this plan. Observed inconvenient outcomes are not an approved exclusion reason.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Inspecting failures and anomalies: the case

02. Follow the deciding steps

03. All four recorded durations

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.